The Structure of Atoms

An atom has a central core called the _____. This is made up of even smaller particles called _____ and _____. The nucleus is surrounded by a cloud of _____.

Each proton has a _____ charge and each electron has a _____ charge, while neutrons are _____.

In a neutral atom, the number of protons is _____ to the number of electrons. This means that the positive charge of the nucleus is _____ to the negative charge of the electrons, so overall the atom is electrically neutral.

Atoms of the same element have the same number of _____ in every atom, but can have different numbers of _____. These are called _____.

Atomic Number

The atomic number of an element is the number of _____ in every atom of that element. For example, a helium atom has 2 protons in its nucleus so it has an atomic number of _____.

Atomic Weight

The atomic weight of an element is the weight of its _____. This is the number of _____ plus the number of _____. The electrons are not counted in the atomic weight because they are very light when compared to the protons and neutrons. A sodium atom has 11 protons and 12 neutrons, so its atomic weight is _____.

In the space below, draw an atom of beryllium. Beryllium has an atomic number of 4 and an atomic weight of 9. Make sure you label the protons, neutrons and electrons in your diagram.

Answer the following questions.

1. An atom of zirconium has 40 protons and 51 neutrons in its nucleus. What is its atomic weight?
2. The atomic number of scandium is 21. Its atomic weight is 45. How many neutrons are there in its nucleus?
3. Manganese has an atomic weight of 55 and its atomic number is 25. How many neutrons does manganese have?
4. Zinc is element number 30. If its atomic weight is 65, how many neutrons are in its nucleus?
5. Draw a labelled diagram of an atom of boron. Boron has an atomic number of 5 and an atomic weight of 11.

6. Use your periodic table to find the information missing below. One row of the table has been done for you.

Element Symbol	Element Name	Atomic Number	Number of Protons	Atomic Weight	Number of Neutrons	Number of Electrons
Pb	Lead	82	82	207	125	82
	Potassium	19		39		
P			15	31		
Ti			22	48		
	Nickel			59		28
Zn		30				
	Tin					
		80				
		56				
	Bromine					
		47				47
				197		